

Derek Phung

B.S. Aerospace Engineering Student · University of California, Irvine

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Education

University of California, Irvine

B.S. in Aerospace Engineering

- eXploration Logistics and Control (XLC) Group (Advisor: Prof. Yuri Shimane)
- GPA: 3.353 / 4.000. Graduate-level coursework: **Space Systems Optimization (A+)**

Irvine, CA

2023.9 - Winter 2027 (expected)

Manuscripts in Preparation

(C1) **Phung, D.**, and Shimane, Y. "High-Fidelity Baseline Design and Station Keeping Analyses for Earth-Moon Vertical Orbits," manuscript in preparation for the 2027 AAS Space Flight Mechanics Meeting.

(J1) Shimane, Y., and **Phung, D.** "HighFidelityEphemerisModel.jl," software manuscript in preparation for the *Journal of Open Source Software*.

Experiences

Undergraduate Researcher / UROP Fellow

University of California, Irvine, XLC Group (Advisor: Prof. Yuri Shimane)

Remote / Irvine, CA

2026.1 - Present

Geometry-Preserving High-Fidelity Baseline Generation and Model Predictive Station Keeping

2026.1 - Present

- Generated eight 5.5-year Earth-Moon L1/L2 vertical-orbit baselines, including six ballistic solutions, and 200 six-month family baselines by mapping CR3BP trajectories into high-fidelity ephemeris dynamics and refining them with sequential convex programming while preserving orbit geometry.
- Developed model predictive station keeping and evaluated robustness through five-year, 100-seed Monte Carlo campaigns across all eight long-duration cases, along with one-year controller/noise sensitivity studies.
- Characterized long-duration orbit behavior through finite-time Lyapunov exponent (FTLE) / stretching analysis, epoch and geometry-constraint studies, and SPICE SPK/BSP validation via direct and propagated-state comparisons.

High-Fidelity Astrodynamics Software Development and Validation

2026.5 - Present

- Implemented Gauss variational equation (GVE) propagation, a native Ephemerides backend, and SPK trajectory utilities/workflows in **HighFidelityEphemerisModel.jl**.
- Validated the GVE and Ephemerides implementations internally against the SPICE backend and performed external HFEM validation against GMAT, Orekit, NBodyPropagator.jl, Nyx/ANISE, and Basilisk using matched physical models and data.
- Built numerical tests and cross-formulation comparisons for Cartesian, modified-equinoctial, Keplerian, and ordinary-equinoctial propagation across shared domains and representative cislunar and lunar-orbit cases.

Ordinary Equinoctial Orbital-Element Support

2026.7 - 2026.8

- Implemented ordinary equinoctial element support in **AstrodynamicsCore.jl**, adding Keplerian, modified-equinoctial, and Cartesian conversion paths, domain-aware validation, round-trip state checks, and API/tutorial documentation.
- Validated conversions across circular, inclined, eccentric, near-equatorial, and near-retrograde cases, including expected failures for singular and non-elliptic states.

Mission Assurance - NASA L'SPACE Mission Concept Academy

NASA L'SPACE Program

Remote

2026.1 - 2026.5

- Performed mission assurance for a multidisciplinary space mission concept, developing risk matrices, mitigation strategies, and hazard controls incorporated into MCR, SRR, MDR, and PDR deliverables.

Fuselage Engineering Team Member

Flying Leatherneck Aviation Museum

Irvine, CA

2025.9 - 2026.4

- Supported reconstruction of a 1918 Curtiss JN-4 "Jenny" aircraft through fabrication, alignment, and SolidWorks modeling of structural and mechanical components from historical drawings.

Pneumatic Autonomous Rover - Controls Lead

University of California, Irvine

Irvine, CA

2025.1 - 2025.3

- Designed an Arduino-based PID controller for autonomous checkpoint navigation, implemented low-pass filtering for magnetometer noise, and tuned closed-loop steering through iterative testing.

Autonomous Rover - Programmer and Researcher

University of California, Irvine

Irvine, CA

2024.1 - 2024.3

- Integrated LiDAR and PixyCam sensing for path alignment and object retrieval, designed the chassis in SolidWorks, and achieved a top-10 completion time in the course competition.

Remote-Controlled Rover - Team Lead and Mechanical Designer

University of California, Irvine

Irvine, CA

2023.9 - 2023.12

- Led a six-member team in rover design and fabrication, including chassis layout and Ackermann steering geometry.

Awards, Programs, & Activities

2023 Fall **Dean's Honor List**, UC Irvine
2024 Spring **Dean's Honor List**, UC Irvine
2025 Fall **Dean's Honor List**, UC Irvine

Miscellaneous

RESEARCH INTERESTS: Astrodynamics; cislunar dynamics; spacecraft guidance, navigation, and control; trajectory optimization; autonomous spacecraft systems; high-fidelity modeling and simulation.

SKILLS: **Programming:** Julia, MATLAB, Python, C++/Arduino, R.
 Astrodynamics: SPICE, JPL ephemerides, CR3BP dynamics, N-body propagation, lunar spherical-harmonic gravity, orbital-element propagation.
 Optimization & Control: sequential convex programming / successive convexification, model predictive control, trajectory optimization, Monte Carlo analysis.
 Software Development: Git, GitHub, scientific software development, numerical validation and testing, LaTeX/Overleaf.
 Engineering: SolidWorks, Siemens NX, Onshape.

LANGUAGES: English (native), Japanese (conversational).